

Enhanced Mathematical Model for UV Degradation of PVC with Tinuvin 770

Parameters

1. $[PVC]$: Concentration of PVC
2. $[R \bullet]$: Concentration of radicals
3. $[S]$: Concentration of Tinuvin 770 (stabilizer)
4. $[P]$: Concentration of stable products formed by radical trapping
5. k_f : Rate constant for radical formation
6. k_q : Rate constant for radical quenching by the stabilizer
7. k_d : Degradation rate constant of PVC in the absence of stabilizer
8. α : Effectiveness factor of Tinuvin 770
9. β : Constant representing the rate of radical formation per unit UV intensity
10. $I(t)$: Intensity of UV radiation as a function of time
11. k_r : Rate constant for radical recombination
12. k_t : Rate constant for radical trapping by the stabilizer

Fundamental Assumptions

1. Radical Formation: The rate of radical formation depends on the UV intensity, $I(t)$, and the concentration of PVC.
2. Radical Quenching: The quenching rate depends on the concentrations of both radicals and Tinuvin 770.
3. PVC Degradation: The degradation rate is influenced by the presence of radicals and the concentration of Tinuvin 770.
4. Radical Recombination: Radical recombination is another pathway for reducing the concentration of radicals.
5. Radical Trapping: Radicals are trapped by Tinuvin 770, forming stable products.

Differential Equations

1. Radical Formation: $\frac{d[R \bullet]}{dt} = \beta I(t)[PVC] - k_q[R \bullet][S] - k_r[R \bullet]^2 - k_t[R \bullet][S]$

2. PVC Degradation: $\frac{d[PVC]}{dt} = -k'_d [PVC]$

where the effective degradation rate constant k'_d is: $k'_d = k_d(1 - \alpha[S])$

3. Stable Product Formation (Radical Trapping): $\frac{d[P]}{dt} = k_t[R \bullet][S]$

4. Stabilizer Concentration: $\frac{d[S]}{dt} = 0$

5. UV Intensity: $I(t) = I_0 \sin^2(\pi t/T)$

where I_0 is the peak UV intensity and T is the period (e.g., 24 hours for a daily cycle).

Solving the Differential Equations

To solve these equations, we can proceed as follows:

1. Radical Concentration:

$$\frac{d[R\bullet]}{dt} = \beta I(t)[PVC] - k_q[R\bullet][S] - k_r[R\bullet]^2 - k_t[R\bullet][S]$$

This is a nonlinear differential equation and typically requires numerical methods for solution.

2. PVC Degradation:

$$\begin{aligned}\frac{d[PVC]}{dt} &= -k'_d[PVC] \\ \frac{d[PVC]}{dt} &= -k_d(1 - \alpha[S])[PVC]\end{aligned}$$

This can be solved analytically:

$$[PVC](t) = [PVC]_0 e^{-k_d(1-\alpha[S])t}$$

Analytical Approach for Radical Concentration

Due to the nonlinearity introduced by the $k_r[R\bullet]^2$ term, the equation for $[R\bullet]$ is complex. We can simplify it by assuming steady-state conditions or using numerical methods. For steady-state conditions ($d[R\bullet]/dt = 0$), we get:

$$0 = \beta I(t)[PVC] - (k_q + k_t)[R\bullet][S] - k_r[R\bullet]^2$$

Solving this quadratic equation for $[R\bullet]$:

$$k_r[R\bullet]^2 + (k_q + k_t)[R\bullet][S] - \beta I(t)[PVC] = 0$$

Using the quadratic formula:

$$[R\bullet] = (-k_q + k_t)[S] \pm \sqrt{\left((k_q + k_t)[S]\right)^2 + \frac{4k_r\beta I(t)[PVC]}{2k_r}}$$

We take the positive root as it represents a physical concentration:

$$[R\bullet] = (-k_q + k_t)[S] + \sqrt{\left((k_q + k_t)[S]\right)^2 + \frac{4k_r\beta I(t)[PVC]}{2k_r}}$$

Summary

- Radical Concentration: $[R\bullet] = (-k_q + k_t)[S] + \sqrt{\left((k_q + k_t)[S]\right)^2 + \frac{4k_r\beta I(t)[PVC]}{2k_r}}$

- PVC Concentration: $[PVC](t) = [PVC]_0 e^{-k_d(1-\alpha[S])t}$

This detailed model accounts for the formation, quenching, and trapping of radicals, radical recombination, and the periodic nature of UV intensity. Solving this model provides insights into the degradation kinetics of PVC and the effectiveness of Tinuvin 770 as a stabilizer under varying UV exposure conditions.